

Research Overview & Questions

The use of generative artificial intelligence tools has grown dramatically since the introduction of OpenAI's ChatGPT in 2022, followed by Anthropic's Claude, Google's Gemini, DeepSeek, and Meta's llama AI. The rapid introduction and implementation of these tools has forced institutions, industries and individuals alike to contend with technology that is simultaneously said to be both transformative and highly overrated. Public discourse surrounding the use of AI tools has oscillated between fears of job loss and skepticism about the efficacy of AI tools in settings where its adoption is both encouraged and forced. Despite the volume of this public discourse, what we know about AI implementation and its use is still preliminary. Since AI's viral entrance into the world at large, there have been widely circulated articles claiming that GPT-4 passed the bar exam (Arredondo 2023). While other credible research conducted by MIT suggests that AI use may contribute to cognitive decline (Kosmyna et al. 2025). At this stage, research has shaped public perception of AI without producing a clear picture as to how students are integrating these tools into their academic lives, or where they are learning to use them. With the promise of AI-driven productivity on one side and the risk of human cognitive decline on the other, this gap in understanding carries real consequence for academic outcomes and the integrity of higher education. At Oregon State University, the institutional response to AI has been to locate decisions on how or if to allow its use in the lap of instructors (AI 2025). This hastened ad-hoc AI policy fails to account for the uneven distribution of AI literacy amongst college students and instructors, which creates conditions where those who possess necessary AI literacy skills hold an advantage over those who do not. This creates an unfair advantage that goes unaddressed in an institution that lacks its own comprehensive AI literacy curriculum to teach its educators how to contend with such disruptive technology. This research seeks to answer the questions:

1. How do undergraduate students at Oregon State University engage with AI tools in the context of academic assignments, and where and how did they learn to use them?
2. What behavioral patterns characterize Human-AI interactions among undergraduate students completing academic tasks, and how do these patterns differ across disciplines and course assignment formats?
3. What institutional and pedagogical responses are most appropriate for addressing inequitable AI literacy among undergraduate students and how can findings inform AI policy and educator training at OSU?

The purpose of this project is to examine the ways undergraduate students at OSU engage with AI tools within academic settings. This study seeks to advance methodological approaches to studying AI-Human interactions by bridging the gap between qualitative ethnographic/reflective methods using participant observation and interviews as well as systematic behavior analysis using both UX based contextual inquiry and ethological methods in the form of ethograms translated from primatology research. By employing these approaches, I aim to create structured behavioral datasets that capture student-AI interactions. An analysis of these interactions will enable me to interpret patterns of AI use as well as their effects on cognition and necessary AI literacy curriculum necessary for future students to come. Specifically, I aim to:

1. Conduct a series of separate AI literacy themed lectures and passive observation sessions within a single undergraduate course at OSU across a 10 week academic term to

document in-class AI engagement through structured observation using both an ethogram and contextual inquiry inspired methods.

2. Conduct passive observational sessions within a separate cohort of undergraduate students enrolled in the same course topic at OSU, without structured AI literacy instruction to document natural AI engagement using ethogram and contextual inquiry inspired methods that serve as a parallel comparison to the approach in the first aim.
3. Assess existing student AI literacy at the beginning of the study and measure change in AI literacy at its conclusion, using findings to identify how exposure to structured AI literacy material and AI engagement within an academic setting affects student understanding and use of AI tools.

Research Structure:

The proposed class for this study will be cultural anthropology (ANTH 210). The motivation to use this class is due to the proximity I have to the instructor overseeing this course, my familiarity to the course content and the amount of sloppy AI use since I and other graduate teaching assistants have been assigned this course. Given that there are several institutional factors shaping the overall layout of the course, the current plan is that there will be two separate cohorts of students split into an AI literacy focused lecture series and a passive AI use lecture series. The lecture series will take place over three different class periods for each cohort and will blend class topics for that week with permitted AI use. Students who choose not to participate will be given an alternate assignment connected to the course topics for that week.

Lecture Series Sessions Overview:

Session 1: Environment & Sustainability		
	Aim 1: Structured AI Literacy (Guided)	Aim 2: Passive AI Use (Observational)
Format	AI Mini Lecture & Guided Activity	Natural Observation Only & Activity
AI Literacy Component	Brief introduction on how AI works & its limitations, how AI is trained, and the realities of AI bias.	None: Students will be introduced to the topic and will be given the option of using AI.
Activity	- Impact of AI on the Environment & Communities. A focus on data centers and AI trainers. - AI Use: Students are allowed to use AI. - AI Use: Students will seek to identify how limitations shape use and identify bias in data produced.	- Impact of AI on the Environment & Communities. A focus on data centers and AI trainers. - AI Use: Students are allowed to use AI.
Observation Focus	Observer will focus on how structured framing of AI development, use and environmental/community impact shape the way students use (or choose not to use) AI.	Observer will focus on how students are using AI (or choosing not to), what AI tools students are using, and to what extent are students using AI.

Session 2: Politics & Power		
	Aim 1: Structured AI Literacy (Guided)	Aim 2: Passive AI Use (Observational)
Format	AI Mini Lecture & Guided Activity	Natural Observation Only & Activity
AI Literacy Component	A brief introduction of human perceptions & use of data, cybernetic theory, and who builds & governs AI tools.	None: Students will be introduced to the week's topic and will be given the option of using AI.
Activity	- Critical comparative analysis of the history of the luddites of the industrial revolution and various modern workers revolutions. Students will seek to understand how different cultural contexts shape power and political responses to disruptive technology. - AI Use: Students are allowed to use AI. - AI Use: Seek to identify how AI describes and possibly minimizes political topics.	Critical comparative analysis of the history of the luddites of the industrial revolution and various modern workers revolutions. Students will seek to understand how different cultural contexts shape power and political responses to disruptive technology. AI Use: Students are allowed to use AI.
Observation Focus	Observer will focus on how structured framing of AI development, use and environmental/community impact shape the way students use AI.	Observer will focus on how students are using AI (or choosing not to), what AI tools students are using, and to what extent are students using AI.

Aim 3: AI Literacy Self Assessment Survey

For both the AI literacy cohort and the passive observational cohort I intend on conducting pre and post study Qualtrics surveys that asks each participant to assess their self perceived AI literacy, perceptions and daily use. The purpose of these surveys will be to track the confidence level of each student and beginning and conclusion of the research in order to better understand the effects of AI literacy curriculum and general exposure to AI use in academic settings. The questions I seek to ask fall under the following categories: Demographic Information (What is your age/gender/major/year(s) in college). Knowledge of AI (How it works, when to use AI, how to use AI). Impacts of AI (Effects of AI on brain, environment, employment). Motivation to use AI (Why do you use AI, What makes you use AI, What makes you use AI)

Intellectual Merit, Broader Impacts & Skill Practice/Development

This research seeks to fill the institutional gap in understanding how academia may begin to contend with AI literacy, AI use, and its impact on the classroom. What isn't well known is how best to implement AI literacy in the classroom, how students are choosing to engage with

AI, what AI tools they are using over others and why, and what resources they are going to in order to educate themselves on how to use AI. By understanding how students are naturally using AI compared to their behaviors with guided AI literacy connected to course content, this research will produce insightful data for a technology that has proved to be disruptive to traditional ways of teaching and learning in higher education.

By using a parallel design of AI literacy lecture sessions and passive AI use sessions, this research will build comparative data that analyzes the efficacy of AI literacy curriculum. An analytical approach using an ethogram, contextual inquiry, and surveys solidifies a mixed methods approach that seeks to identify behavioral patterns of student AI use. This will create actionable qualitative and quantitative data to help educators and institutions inform curriculum and AI policies as AI tech becomes more pervasive. Beyond the classroom, this research may meaningfully impact how students utilize AI tools as they transition into the workforce. By understanding how students engage with AI and respond to AI literacy instruction, this research will show how students form applied understandings of AI as the knowledge of how to use AI becomes more pervasive in society, this research may also shape expectations of how work places expects graduates to possess functional and critical AI competency.

My background in computer programming languages like Python, R, and C++ coupled with my anthropological background puts me in the unique position of understanding the cross section of technology and its effects on academia, students and culture guiding modern education spaces. There is extensive research being done on AI innovation, I feel that my perspective and research approach utilizes user experience and anthropological methods needed for understanding how AI affects our education system at the individual classroom level.

This research will develop several applied research skills. Designing and delivering AI literacy instruction that is integrated with anthropological curriculum builds on pedagogical and translational skills that communicate complex technical concepts within a human focused research design. Conducting participant observation and contextual inquiry in active classroom settings will strengthen ethnographic fieldwork skills in a non-traditional research environment. Developing and administering pre- and post-study surveys will provide direct experience in human subject data collection. Lastly, compiling and analyzing ethogram data will build proficiency in systematic behavioral observation and quantitative coding methods translated from UX and primatology research.

Budget

Item	Cost
Participant Incentives	\$1,100
Total	\$1,100

Participant Incentives

The sample size for this study will be 82 students, assuming that each class has a roster of about 41 students each. In exchange for their participation and submission of data for the study, they will be entered into a raffle for a \$25 gift card of which 44 students will be chosen, 22 from each cohort. By giving out incentives through a raffle, I avoid creating a coercive environment where students feel forced to participate in the study.

References

- Arredondo, Pablo. 2023. "GPT-4 Passes the Bar Exam: What That Means for Artificial Intelligence Tools in the Legal Profession." *Stanford Law School Blog*, April 19. <https://law.stanford.edu/2023/04/19/gpt-4-passes-the-bar-exam-what-that-means-for-artificial-intelligence-tools-in-the-legal-industry/>
- Kosmyna, Nataliya., Eugene Hauptmann, Ye Tong Yuan, et al. 2025. "Your Brian on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task." *MIT Media Lab*. <https://doi.org/10.48550/arXiv.2506.08872>